Eito Shinokura

Lead AI/Data Engineer

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Summary

Lead AI/Data Engineer with over 6 years of experience architecting and deploying advanced AI systems across enterprise data science, automation, and intelligent agent platforms. Specialized in Generative AI, MLOps, and scalable backend infrastructure, having evolved from traditional machine learning and data analytics to cutting-edge LLM-based AI architectures. Successfully delivered production-grade AI solutions that transformed business operations for Fortune 500 companies and scale-ups, with deep expertise in backend development using Python and modern cloud technologies.

Professional Experience

Lead AI/Data Engineer
Foundation EGI, Los Altos, CA, US
March 2025–Present

- Architected multi-agent LLM systems using LangGraph, CrewAI, and AutoGen to automate finance, HR, and operations workflows, reducing operational costs by 35% for Fortune 500 clients and enabling 3x team productivity gains
- Designed and implemented RAG pipelines integrating LlamaIndex with Pinecone and Weaviate vector databases, achieving 92% accuracy in semantic search across 10M+ enterprise documents
- Engineered production-grade AI agents using GPT-4, Claude 3.5 Sonnet, and Gemini Pro with advanced prompt engineering and function calling, achieving an 88% automation success rate in production environments
- Developed real-time agent orchestration systems using LangChain and LangSmith, processing 5M+ enterprise workflows monthly with sub-second latency across multiple departments
- Established comprehensive MLOps infrastructure using Azure ML, MLflow, and Weights & Biases with CI/CD pipelines, reducing model deployment cycles from 3 weeks to 4 hours
- Constructed scalable backend microservices using FastAPI and Azure Functions with async processing, handling 8M+ daily automation requests with 99.9% uptime
- Deployed fine-tuned LLMs using vLLM and Azure OpenAI on Kubernetes clusters with auto-scaling, reducing inference costs by 45% while maintaining high throughput
- Integrated comprehensive monitoring and cost optimization using Helicone and LangSmith, reducing LLM operational costs by 40% through token tracking and intelligent caching
- Spearheaded technical architecture discussions with 15+ Fortune 500 clients, translating complex requirements into scalable AI solutions and driving successful AI adoption across organizations

Senior AI Engineer
Tenchijin, Minato, Tokyo, Japan
April 2022–February 2025

- Produced intelligent document understanding systems using BERT, RoBERTa, and LayoutLM, achieving 94% extraction accuracy and processing 500K+ business documents monthly for European enterprise clients
- Orchestrated NLP pipelines for multi-language sentiment analysis and entity recognition using spaCy and Transformers, supporting 4 languages with 91% accuracy, exceeding industry benchmarks by 8%
- Built question-answering systems using BERT and Sentence-BERT with FAISS, enabling sub-second queries across 5M+ documents with 89% answer relevance score
- Crafted and deployed recommendation engines using collaborative filtering and neural networks with PyTorch, improving client conversion rates by 28% and driving $2M+ in additional revenue
- Pioneered MLOps best practices using MLflow, DVC, and Kubeflow on AWS, implementing automated retraining pipelines that improved model performance by 15% and reduced time-to-production by 60%

- Architected backend services using FastAPI and Django with PostgreSQL and Redis, building scalable APIs supporting 10K+ requests/minute for real-time NLP inference
- Optimized model inference using ONNX Runtime and TorchServe on AWS, achieving 3.5x speedup and 50% cost reduction while maintaining 99.2% model accuracy
- Partnered with 20+ enterprise clients on-site and remotely, formulated end-to-end ML solutions, and provided ongoing support, ensuring a 95% client retention rate

Machine Learning Engineer
Citadel AI, Shibuya-ku, Tokyo, Japan
April 2020–March 2022

- Delivered computer vision systems using CNNs with ResNet, EfficientNet, and YOLO, achieving 96% detection accuracy and reducing manual inspection time by 75% for manufacturing clients
- Engineered NLP solutions for text classification and named entity recognition using LSTM, GRU, and BERT models, processing 100K+ documents monthly with 87% F1-score
- Designed predictive analytics models using XGBoost, LightGBM, and Random Forests for demand forecasting, improving prediction accuracy by 32% and reducing inventory costs by $800K annually
- Analyzed large-scale datasets using Apache Spark, Pandas, and SQL on AWS EMR, processing 50TB+ data monthly and uncovering insights that drove $1.5M in cost savings
- Executed MLOps workflows using Docker, Jenkins, and Kubernetes on GCP and AWS, establishing CI/CD pipelines that reduced model deployment time by 70% and improved reliability to 99.5%
- Developed backend APIs using Flask and Django REST Framework with MySQL and MongoDB, creating microservices supporting 5K+ requests/minute for ML model serving
- Facilitated technical workshops for 10+ client teams, devised solution architectures, and mentored 5 junior engineers in data science and software development best practices

Education

Master's Degree | April 2020 - March 2022 | The University of Tokyo **Tokyo, Japan**
Bachelor's Degree | April 2016 - March 2020 | Tokyo Institute of Technology **Tokyo, Japan**

Skills & Abilities

- Programming Languages: Python, Java, JavaScript, C/C++, R, SQL, Go, Scala
- LLM & GenAI Frameworks: LangChain, LangGraph, LangSmith, LlamaIndex, CrewAI, AutoGen, Ollama, LiteLLM, Helicone, DSPy, Guidance, Semantic Kernel
- ML/DL Frameworks: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Transformers, Hugging Face, spaCy, NLTK, Keras, FastAI, Sentence-Transformers
- MLOps & Tools: MLflow, Weights & Biases, DVC, Kubeflow, Apache Airflow, Docker, Kubernetes, Jenkins, GitHub Actions, TensorBoard, Prefect
- Optimization & Deployment: vLLM, ONNX, TensorRT, TorchServe, TensorFlow Serving, Triton Inference Server, Model Quantization (GPTQ, AWQ, GGUF)
- Cloud Platforms: Microsoft Azure (Azure ML, OpenAI Service, Functions, AKS), AWS (SageMaker, Lambda, EKS, EC2, EMR), Google Cloud Platform (GCP)
- Vector Databases: Pinecone, Weaviate, Qdrant, FAISS, Chroma, Milvus, Elasticsearch
- Databases: PostgreSQL, MongoDB, MySQL, Redis, SQLite, DynamoDB, Cassandra
- Backend Frameworks: FastAPI, Flask, Django, Django REST Framework, Node.js, Express.js, Celery, SQLAlchemy, Pydantic
- Data Processing: Pandas, NumPy, Apache Spark, Apache Kafka, Dask, Polars, Apache Beam, PySpark